Chapter-4

Joint probability function-

$P[x=x_i, Y=Y_j]$ be the joint probability function of taking $x=x_i$ and $Y=Y_j$.

$P[x=x_i, Y=Y_j]$ for all x_i and Y_j is known as joint probability function. if

i) $P[x=x_i, Y=Y_j] \geq 0$ for all x_i, Y_i

2) $\sum_i \sum_j P(x=x_i, Y=Y_j) = 1$

$\Rightarrow$ X, Y - CRV

$f(x,y)$ will be joint probability function if.

i) $f(x,y) \geq 0$

ii) $\int_{x=-\infty}^{\infty} \int_{Y=-\infty}^{\infty} f(x,y)\, dx\, dy = 1$

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Marginal density function-

$$g(x) = \int_{Y=-\infty}^{Y=\infty} f(x,Y)\, dY$$

$$h(Y) = \int_{x=-\infty}^{x=\infty} f(x,Y)\, dx$$

} X, Y continuous RV

D. $$g(x) = \sum_{j=1}^{n} p(x_i, Y_j)$$

$$h(Y) = \sum_{i=1}^{m} p(x_i, Y_j)$$

} X, Y DRV

$$P[x = x_i,\ Y = Y_i) \rightarrow \begin{matrix} i = 1 \text{ --- } m \\ j = 1 \text{ --- } n \end{matrix}$$

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Chapter-4

Marginal & distribution
In discrete case -
Marginal distribution of X_i

$$P_1(x_1) = \sum_{x_2} P(x_1, x_2) \quad \text{for all } x_1$$

$X_2 \rightarrow$

$$P_2(x_2) = \sum_{x_1} P(x_1, x_2) \quad \text{for all } x_2$$

Continuous —

$$f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)\, dx_2$$

Ex. 4.6)

$$f(x_1, y_1) = \frac{1}{500} \qquad 0 < x_1 < 0.25;\ 0 < x_2 < 2000$$

$$f_1(x_1) = \int_0^{2000} \frac{1}{500}\, dx_2$$

$$= \left[\frac{x_2}{500}\right]_0^{2000}$$

$$= \frac{2000}{500} = 4$$

$$\therefore f_1(x_1) = \begin{cases} 4 & ; 0 < x_1 < 0.25 \\ 0 & ; \text{other} \end{cases}$$

$$f_2(x_2) = \int_0^{0.25} \frac{1}{500}\, dx_1$$

$$= \frac{0.25}{500}$$

$$= \begin{cases} \frac{1}{2000} & ; 0 < x_2 < 2000 \\ 0 & ; \text{other} \end{cases}$$

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Ex 4.7)

$$f(x_1, x_2) = \begin{cases} 6x_1 & 0 < x_1 < x_2 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Note limit

$$\begin{bmatrix} 0 < x_1 < x_2 \\ x_1 < x_2 < 1 \end{bmatrix}$$

marginal of x_1

$$f_1(x_1) = \int_{x_1}^{1} 6x_1 \, dx_2$$

$$= 6x_1 [x_2]_{x_1}^{1}$$

$$= 6x_1 (1 - x_1) \quad \text{for} \quad 0 < x_1 < 1$$

Marginal of x_2

$$f_2(x_2) = \int_0^{x_2} 6x_1 \, dx_1$$

$$= 6 \left[\frac{x_1^2}{2} \right]_0^{x_2}$$

$$= 3 x_2^2 \quad \text{for} \quad 0 < x_2 < 1$$

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$[X_1, X_2]$ continuous

$$E(X_1) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 f_1(x_1, x_2)\, dx_2\, dx_1$$

$$E(X_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_2 f_2(x_1, x_2)\, dx_1\, dx_2$$

$$V(X_1) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1^2 f(x_1, x_2)\, dx_2\, dx_1 - \mu_1^2$$

$$V(X_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_2^2 f(x_1, x_2)\, dx_1\, dx_2 - \mu_2^2$$

4.9)

$$f(x_1, x_2) = \begin{cases} \frac{1}{500} & 0 \le x_1 < 0.25 \\ & 0 \le x_2 \le 2000 \\ 0 & \text{other} \end{cases}$$

$$f_1(x_1) = \begin{cases} 4 & ; 0 \le x < 0.25 \\ 0 & ; \text{other} \end{cases}$$

$$f_2(x_2) = \begin{cases} \frac{1}{2000} & ; 0 \le x_2 \le 2000 \\ 0 & ; \text{other} \end{cases}$$

$$E(X_1) = \mu_1 = \int_{-\infty}^{\infty} x_1 f_1(x_1)\, dx_1 \quad ***$$

$$E(X_2) = \mu_2 = \int_{-\infty}^{\infty} x_2 f_2(x_2)\, dx_2$$

$$V(X_1) = \int_{-\infty}^{\infty} x_1^2 f_1(x_1)\, dx_1 - \mu_1^2$$

$$V(X_2) = \int_{-\infty}^{\infty} x_2^2 f_2(x_2)\, dx_2 - \mu_2^2$$

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~~f(x,y~~ $f(x,y) \rightarrow x$ CRV

$$g(x) = \int_Y f(x,y)\,dy$$

$$h(y) = \int_x f(x,y)\,dx$$

$$E(x) = \mu_1 = \int_x x\, g(x)\,dx$$

$$E(y) = \mu_2 = \int_y y\, h(y)\,dy$$

$$\text{var}(x) = \int_x x^2 g(x)\,dx - \mu_1^2$$

$$\text{var}(y) = \int_y y^2 h(y)\,dy - \mu_2^2$$

Conditional prob -

$$f_{x|y}(x) = \frac{f(x,y)}{h(y)}$$

$$f_{y|x}(y) = \frac{f(x,y)}{g(x)}$$

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ex → 4.8, 4.9 $f(x_1, x_2) = \frac{1}{500}$ $0 \le x_1 \le 0.25$,

$0 \le x_2 \le 2000$

$$f_1(x_1) = \int_0^{2000} \frac{1}{500}\, dx_2$$

$$= \frac{1}{500}(2000 - 0)$$

$$= 4; \quad 0 \le x_1 \le 0.25$$

$$f_2(x_2) = \int_0^{0.25} \frac{1}{500}\, dx_1$$

$$= \frac{0.25}{500}$$

$$= \frac{1}{2000} \quad 0 \le x_2 \le 2000$$

Mean,

$$\mu_1 = E(x_1) = \int_0^{0.25} x_1 f_1(x_1)\, dx_1$$

$$= \int_0^{0.25} x_1 4\, dx_1$$

$$= 4 \left(\frac{x_1^2}{2}\right)_0^{0.25}$$

$$= 2 \times 0.0625$$

$$= 0.125$$

$$V(X_1) = \int_0^{0.25} x_1^2 f(x_1) dx_1 - \mu_1^2$$

$$= \int_0^{0.25} x_1^2 4\, dx_1 - 0.125^2$$

$$= 4\left[\frac{x_1^3}{3}\right]_0^{0.25} - 0.015625$$

$$= 5.21 \times 10^{-3}$$

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Lecture - 10 (cont)

chap - 4

7/11/21

Conditional distribution.

$$P_{x_2|x_1}(x_2) = \frac{P(x_1, x_2)}{P_1(x_1)} \quad \text{for all } x_1, x_2$$

$$P_{x_1|x_2}(x_1) = \frac{P(x_1, x_2)}{P_2(x_2)} \quad ''$$

where $P_1(x_1) > 0$, $P_2(x_2) > 0$

Ex 4-11

$$f(x_1, y_1) = x_1^2 + \frac{x_1 x_2}{3} \quad 0 < x_1 \le 1, \ 0 \le x_2 \le 2$$

$$= 0 \quad \text{other}$$

a) $P_{x_2|0}(x_2) = \dfrac{P(0, x_2)}{P(0)}$

Conditional Expectation

$$E(X_1|X_2) = \sum_{x_i} x_1 P_{x_1|x_2}(x_1)$$

$$= \int_{-\infty}^{\infty} x_1 f_{x_1|x_2}(x_2)\, dx_1$$

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x f_x(x)\, dx$$

$$E(Y) = \mu_Y = \int_{-\infty}^{\infty} y f_Y(Y)\, dy$$

$$V(X) = \int_{-\infty}^{\infty} x^2 f_x(x)\, dx - \mu_X^2$$

$$V(Y) = \int_{-\infty}^{\infty} y^2 f_Y(Y)\, dy - \mu_Y^2$$

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Ex→ A bank operate both drive up facility and a walk-up window. On a random selected day X = the proportion of time that the drive up facility is in use (at least one customer is served or waiting to be served) and Y = proportion of time that walk up window is in use. Then the set of possible values for (X, Y) is rectangle

$$D = \{(x, y) ; 0 \le x \le 1, 0 \le y \le 1\}$$

$$f_{xy}(x, y) = \frac{6}{5}(x + y^2) \quad 0 \le x \le 1, 0 \le y \le 1$$

i) Verify if $f_{xy}(x, y)$ is legitimate pdf

⇒ $f_{xy}(x, y)$ will be pdf if

$$\int_0^1 \int_0^1 f_{xy}(x, y)\, dx\, dy = 1$$

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$$= \int_0^1 \int_0^1 \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_0^1 \left[\frac{x^2}{2} + xy^2\right]_0^1 dy$$

$$= \frac{6}{5}\int_0^1 \left(\frac{1}{2} + y^2\right)dy$$

$$= \frac{6}{5}\left[\frac{xy}{2} + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left(\frac{1}{2} + \frac{1}{3}\right)$$

$$= \frac{6}{5} \times \frac{5}{6} = 1$$

—o—

n) What is the probability that neither facility is busy more than $\frac{1}{4}$ time

$$P\left(x \le \frac{1}{4}, y \le \frac{1}{4}\right)$$

$$= \int_0^{1/4} \int_0^{1/4} \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_0^{1/4} \left[\frac{x^2}{2} + xy^2\right]_0^{1/4} dy$$

$$= \frac{6}{5}\int_0^{1/4} \left(\frac{1}{32} + \frac{y^2}{4}\right)dy$$

$$= \frac{6}{5}\left[\frac{y}{32} + \frac{y^3}{12}\right]_0^{1/4}$$

$$= \frac{6}{5}\left(\frac{1}{32\times 4} + \frac{1}{768}\right)$$

$$= \frac{6}{5}\left(\frac{6+1}{768}\right)$$

$$= \frac{7}{640}$$

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iii) Marginal Pdf of X,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\, dy$$

$$= \frac{6}{5}\left[xy + \frac{y^3}{3} \right]_0^1$$

$$= \frac{6}{5}\left(x + \frac{1}{3}\right)$$

$$= \frac{6x}{5} + \frac{2}{5} \quad ; \quad 0 \le x \le 1$$

Marginal Pdf of Y

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dx$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\, dx$$

$$= \frac{6}{5}\left(\frac{x^2}{2} + xy^2\right)_0^1$$

$$= \frac{6}{5}\left(\frac{1}{2} + y^2\right)$$

$$= \frac{3}{5} + \frac{6y^2}{5} \quad ; \quad 0 \le y \le 1$$

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...ased for ... not determined
...tween is ... only different

IV) Evalute conditional Pdf of Y given $x=0.8$

~~$f_{Y|x}(Y) = \frac{f_{xy}(x,y)}{f_x(x)}$~~

$$f_{Y|x}(Y) = \frac{f_{xy}(x,y)}{f_x(x)}$$

$$= \frac{\frac{6}{5}(x+y^2)}{\frac{6x}{5}+\frac{2}{5}}$$

$$= \frac{x+y^2}{(x+\frac{1}{3})} \qquad 0 \le x \le 1, \quad 0 \le y \le 1$$

$$= \frac{0.8+y^2}{0.8+\frac{1}{3}}$$

$$= \frac{3}{17}(4+5y^2) \qquad 0 \le y \le 1$$

$$P(Y>0 \mid x=0.8) = \int_0^\infty f_{Y|x}(Y)\, dy$$

$$= \int_0^1 \frac{3}{17}(4+5y^2)\, dy$$

$$= \frac{3}{17}\left[4y + \frac{5y^3}{3}\right]_0^1$$

$$= \frac{3}{17}\left(4+\frac{5}{3}\right)$$

$$= \frac{3}{17} \times \frac{17}{3}$$

$$= 1$$

—o—

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